

1. Original Case Process

The original SimuDec study was developed for a close-hole drilling-reaming process of a tractor lever component manufactured from grey cast iron FC250 in a Thai manufacturing context. The process involved producing a starting hole on a milling machine using a dedicated jig, followed by reaming to achieve the required dimensional accuracy, roundness, and surface finish.

In the case study, incomplete reaming referred to a reamed hole that did not satisfy the required hole specification or that exhibited a characteristic reaming-related quality defect. SimuDec was intended to support the engineering decision stage after an unacceptable or abnormal reaming outcome had been observed, rather than merely to determine whether a defect was present.

1.1 Target Hole and Quality Requirements

Item	Specification
Component material	Grey cast iron FC250
Target hole diameter	10 mm
Target hole depth	30 mm
Diameter tolerance	± 0.01 mm
Depth tolerance	± 0.05 mm
Roundness requirement	< 0.01 mm
Surface roughness requirement	$< Ra\ 3.2\ \mu m$

1.2 Defect Manifestations Presented as Context

Across the original scenario set, respondents were presented with reaming-related quality manifestations including poor surface finish, poor roundness, chatter marks, oversize holes, undersize holes, scoring, and taper. The defect manifestation was used as contextual information during engineering judgement and was not stored as a separate predictor in the archived machine-learning training dataset.

2. Taguchi L18 Scenario Design

The questionnaire scenarios were generated using a Taguchi L18 orthogonal array containing one two-level factor (coolant condition) and seven three-level factors. Tool life was controlled at 1,000

h. The factor levels reported for the original case study are summarised in Table S-A1.

Table S-A1. Factors and levels used in the original SimuDec scenario design

Factor	Level 1	Level 2	Level 3
Coolant condition	On	Off	-
Drilling speed (rpm)	600	800	1,000
Reaming speed (rpm)	200	500	700
Feed rate (mm/rev)	0.10	0.15	0.20
Tool usage time (h)	10	500	900
Pilot-hole diameter (mm)	9.7	9.8	9.9
Runout (mm)	0.01	0.03	0.06
Tool overhang (mm)	20	40	60

Note: For model training, tool usage was represented as a tool-wear fraction of 0.01, 0.50, and 0.90 for the three tool-usage levels.

2.1 Archived S01-S18 Machining-Condition Matrix

Table S-A2 reproduces the 18 machining-condition combinations retained in the archived SimuDec scenario implementation. Expected root-cause and corrective-action labels are intentionally omitted because the original questionnaire required respondents to provide these decisions.

Table S-A2. Taguchi L18 machining-condition matrix

ID	Coolant	Drill (rpm)	Ream (rpm)	Feed (mm/rev)	Tool use (h)	Pilot hole (mm)	Runout (mm)	Overhang (mm)
S01	On	600	200	0.10	10	9.7	0.01	20
S02	On	600	200	0.15	500	9.8	0.03	40
S03	On	600	200	0.20	900	9.9	0.06	60
S04	On	800	500	0.10	10	9.8	0.03	60
S05	On	800	500	0.15	500	9.9	0.06	20
S06	On	800	500	0.20	900	9.7	0.01	40
S07	On	1000	700	0.10	500	9.7	0.06	40
S08	On	1000	700	0.15	900	9.8	0.01	60
S09	On	1000	700	0.20	10	9.9	0.03	20
S10	Off	600	200	0.10	900	9.9	0.03	40
S11	Off	600	200	0.15	10	9.7	0.06	60
S12	Off	600	200	0.20	500	9.8	0.01	20
S13	Off	800	500	0.10	500	9.9	0.01	60
S14	Off	800	500	0.15	900	9.7	0.03	20
S15	Off	800	500	0.20	10	9.8	0.06	40
S16	Off	1000	700	0.10	900	9.8	0.06	20
S17	Off	1000	700	0.15	10	9.9	0.01	40
S18	Off	1000	700	0.20	500	9.7	0.03	60

3. Scenario-Based Questionnaire Structure

Expert knowledge was collected using a closed-ended, scenario-based questionnaire. For each of the 18 scenarios, respondents reviewed the machining conditions together with the associated defect manifestation and recorded the decision information listed in Table S-A3.

Table S-A3. Documented structure of the original SimuDec questionnaire

Item	Response format	Purpose
Primary root cause	Single selection	Select the most likely principal cause of the observed reaming problem.
Secondary root cause	Optional selection	Identify an additional plausible contributing cause, where applicable.
Production decision	Single selection	Record the production-related decision associated with the scenario.
Primary corrective action	Single selection	Select the principal action to address the identified problem.
Secondary corrective action	Optional selection	Identify an additional corrective action, where applicable.
Confidence	Rating	Indicate confidence in the selected decision. The archived results report this on a five-point scale.
Expected effectiveness	Rating	Assess the expected effectiveness of the proposed corrective action.
Urgency	Rating	Assess the urgency of implementing the corrective action.

3.1 Root-Cause Response Categories

The archived SimuDec study used eight primary root-cause categories. The wording below follows the archived study terminology.

Code	Root-cause category
A1	Tool wear / tool condition
A2	Chip packing / chip evacuation
A3	Inappropriate spindle speed
A4	Inappropriate feed rate
A5	Coolant not used

A6	Improper pilot hole
A7	Runout / alignment / work holding
A8	Excessive tool overhang

3.2 Corrective-Action Response Descriptions

The archived materials retain the following eight corrective-action descriptions. They are listed without reproducing conflicting legacy code numbers because the archived manuscript contains an indexing inconsistency in its corrective-action summary table. The current Literature-Grounded SimuDec-RAG paper uses the harmonised C1-C8 coding shown in its main taxonomy table.

No.	Corrective-action description
1	Replace or regrind the reamer
2	Adjust the spindle speed
3	Adjust the feed rate
4	Use / correct coolant or lubrication condition
5	Modify the pilot-hole size or drilling process
6	Correct runout, alignment, or work holding
7	Reduce tool overhang
8	Improve chip evacuation

Archival note: The archived manuscript states that corrective-action classes C1-C8 were used, but its label-distribution table displays “Reduce tool overhang” as C8 and “Improve chip evacuation” as C9, with no C7. A later archived SimuDec implementation uses an eight-action C1-C8 scheme. This supplementary document therefore preserves the action descriptions while avoiding a misleading reproduction of the inconsistent legacy numbering.

4. Questionnaire Template Used for Each Scenario

The following template reflects the documented structure of the original instrument. It is provided to show how each Taguchi scenario was presented for engineering judgement. Exact verbal anchors for all rating scales and the complete production-decision response options were not preserved in the archived manuscript; therefore, they are not reconstructed here.

Scenario ID: [S01-S18]	
Machining condition	Coolant; drilling speed; reaming speed; feed rate; tool usage; pilot-hole diameter; runout; tool overhang
Observed defect manifestation	Scenario-specific quality manifestation presented to the respondent
Q1. Primary root cause	Select one primary root-cause category
Q2. Secondary root cause	Optional additional cause
Q3. Production decision	Select the applicable production decision
Q4. Primary corrective action	Select one primary corrective action
Q5. Secondary corrective action	Optional additional corrective action
Q6. Confidence	Rating; archived analysis reports a five-point confidence scale
Q7. Expected effectiveness	Rating
Q8. Urgency	Rating

5. Respondents and Archived Dataset

A total of 154 engineers and operators within the case company completed all 18 scenarios, producing 2,772 complete respondent-scenario records. The archived model-training dataset retained eight machining-condition predictors: coolant condition, drilling speed, reaming speed, feed rate, tool-wear fraction, pilot-hole diameter, runout, and tool overhang.

The scenario-specific defect manifestation was presented as contextual information during the questionnaire but was not encoded as a separate predictor in the archived model-training dataset. Primary root-cause and primary corrective-action selections were used as the two machine-learning target variables.

The archived analysis dataset did not retain verified role-specific respondent counts or years-of-experience variables. Formal Content Validity Index (CVI) and psychometric reliability coefficients were also not available for the questionnaire. Accordingly, the original responses should be interpreted as process-specific engineering judgement rather than as a formally validated psychometric instrument.

6. Relationship to the Literature-Grounded SimuDec-RAG Study

The original SimuDec case, scenarios, and questionnaire provided the historical basis for the root-cause and corrective-action structure subsequently used in the Literature-Grounded SimuDec-RAG study. In the current work, the category descriptions were refined to accommodate a broader range of reaming mechanisms and the decision process was extended from closed scenario-derived knowledge to retrieval from peer-reviewed manufacturing literature. The current taxonomy and experimental protocol are defined in the main manuscript.

Source Note

This supplementary document was compiled from the archived SimuDec manuscript, “Training Machine Learning with Simulated Scenarios for Defect-to-Decision Mapping in Manufacturing: The SimuDec Framework”, together with the archived S01-S18 scenario implementation. No unverified respondent demographics, questionnaire reliability statistics, or scenario-level defect descriptions have been added.